

Poisson Regression

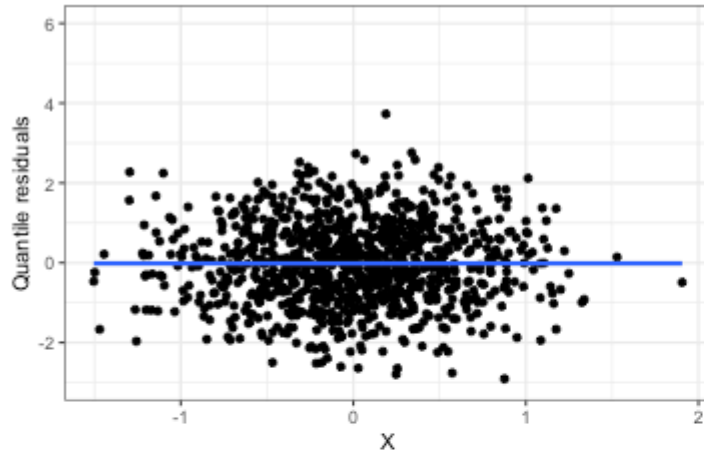
Class activity: Poisson regression assumptions

https://sta214-f25.github.io/class_activities/ca_21.html

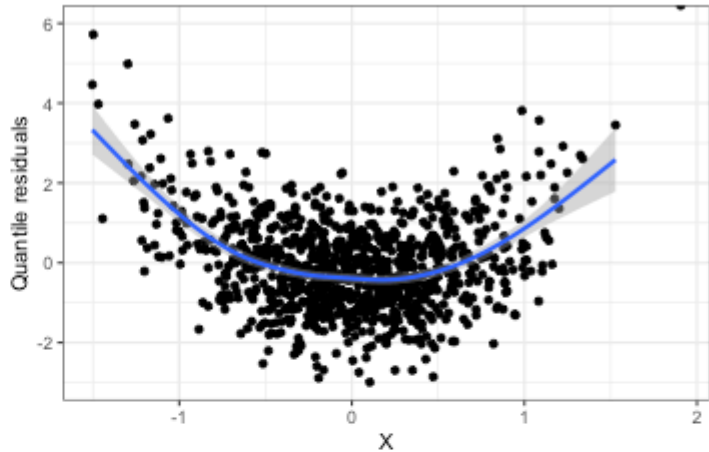
Work on the class activity, then we will discuss as a class.

Class activity: Poisson regression assumptions

Poisson data, shape assumption satisfied

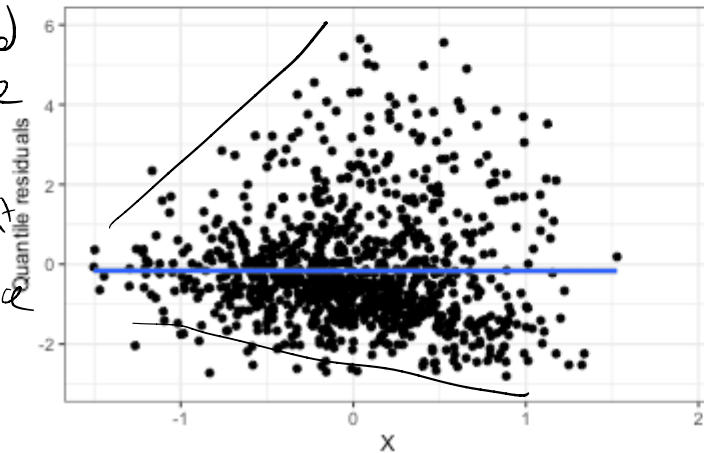


Poisson data, shape assumption violated



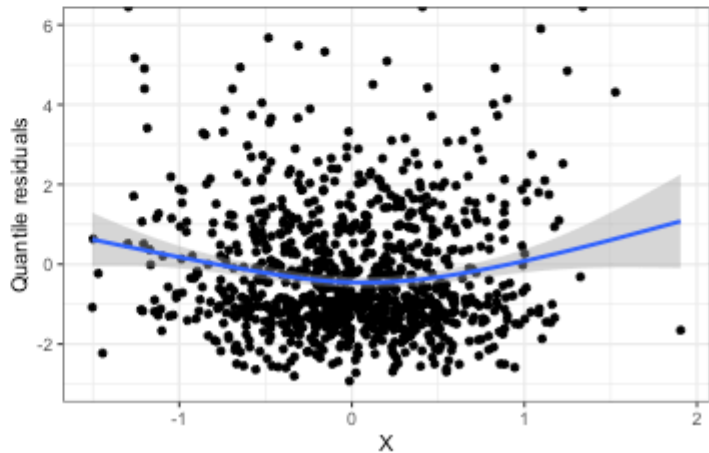
pattern
(U-shape)

Non-Poisson data, shape assumption satisfied



increased
variance
non-
constant
variance

Non-Poisson data, shape assumption violated



pattern
increased
variance

Using quantile residual plots

All assumptions are satisfied:

- + Residuals are randomly scattered around 0, with no clear pattern
- + Residuals have constant variance
- + Residuals are mostly between -2 and 2

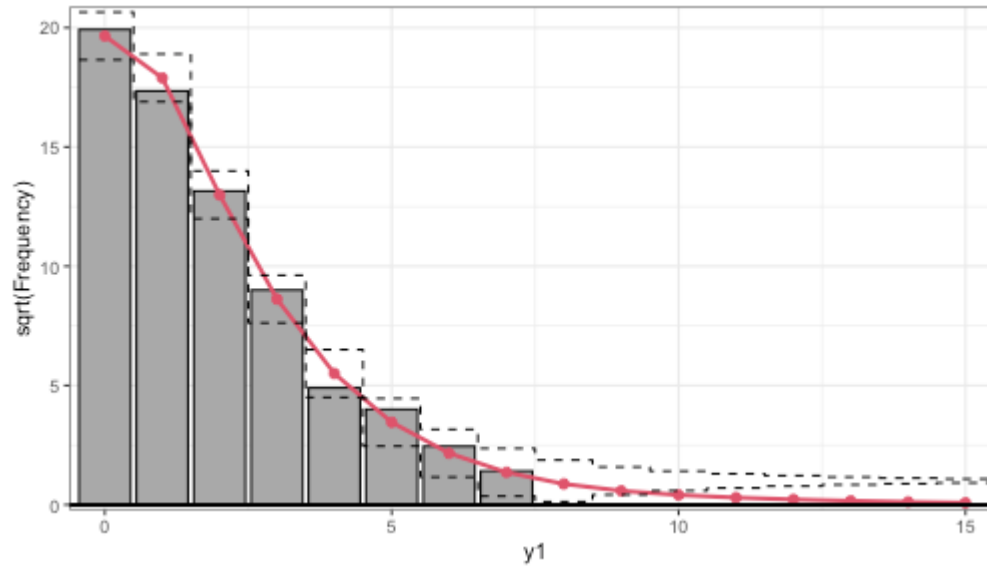
If you see a pattern in the residuals:

- + Shape assumption may not be satisfied

If residuals have non-constant variance, and/or variance is too large:

- + Distribution assumption may not be satisfied

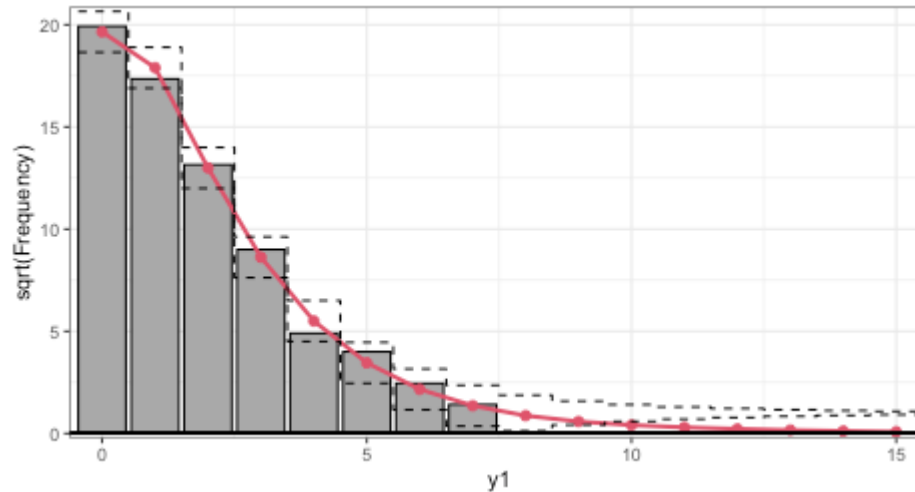
Rootograms



How are rootograms created?

$$Y \sim \text{Poisson}(\lambda)$$

$$P(Y=0) = \frac{\lambda^0 e^{-\lambda}}{0!} = \lambda^0 e^{-\lambda}$$



```
sqrt(sum(y1 == 0))
```

```
## [1] 19.92486
```

$\uparrow$ # times $Y=0$

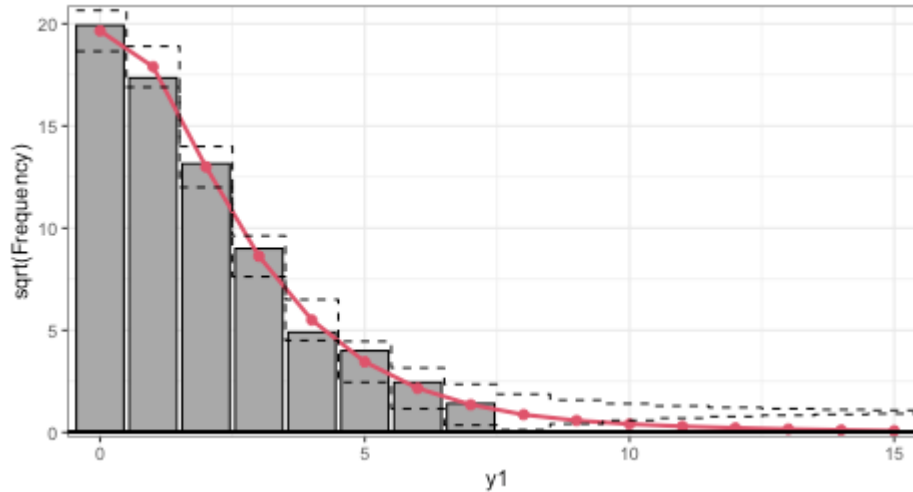
```
lambda_hat <- m1$fitted.values  
sqrt(sum(lambda_hat^0 * exp(-lambda_hat)))
```

```
## [1] 19.65352
```

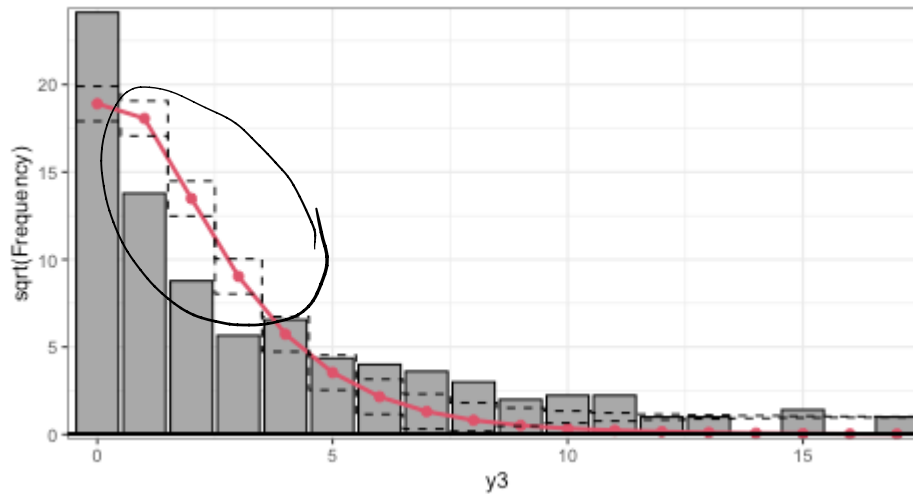
$\hat{P}(Y_i = 0)$

Examining rootograms

model
assumptions
are satisfied



model
assumptions
are not
satisfied



Examining rootograms

If all assumptions are satisfied (i.e., the model is a good fit to the data):

- + The histogram bars will be close to the predictions (in red) from the fitted model

If the model is not a good fit to the data (wrong shape and/or distribution):

- + The predictions will not match the observed frequencies

Another dataset

A concerned parent asks us to investigate crime rates on college campuses. We have access to data on 81 different colleges and universities in the US, including the following variables:

- + type: college (C) or university (U)
- + nv: the number of crimes for that institution in the given year
- + enroll1000: the number of enrolled students, in thousands
- + region: region of the US C = Central, MW = Midwest, NE = Northeast, SE = Southeast, SW = Southwest, and W = West)

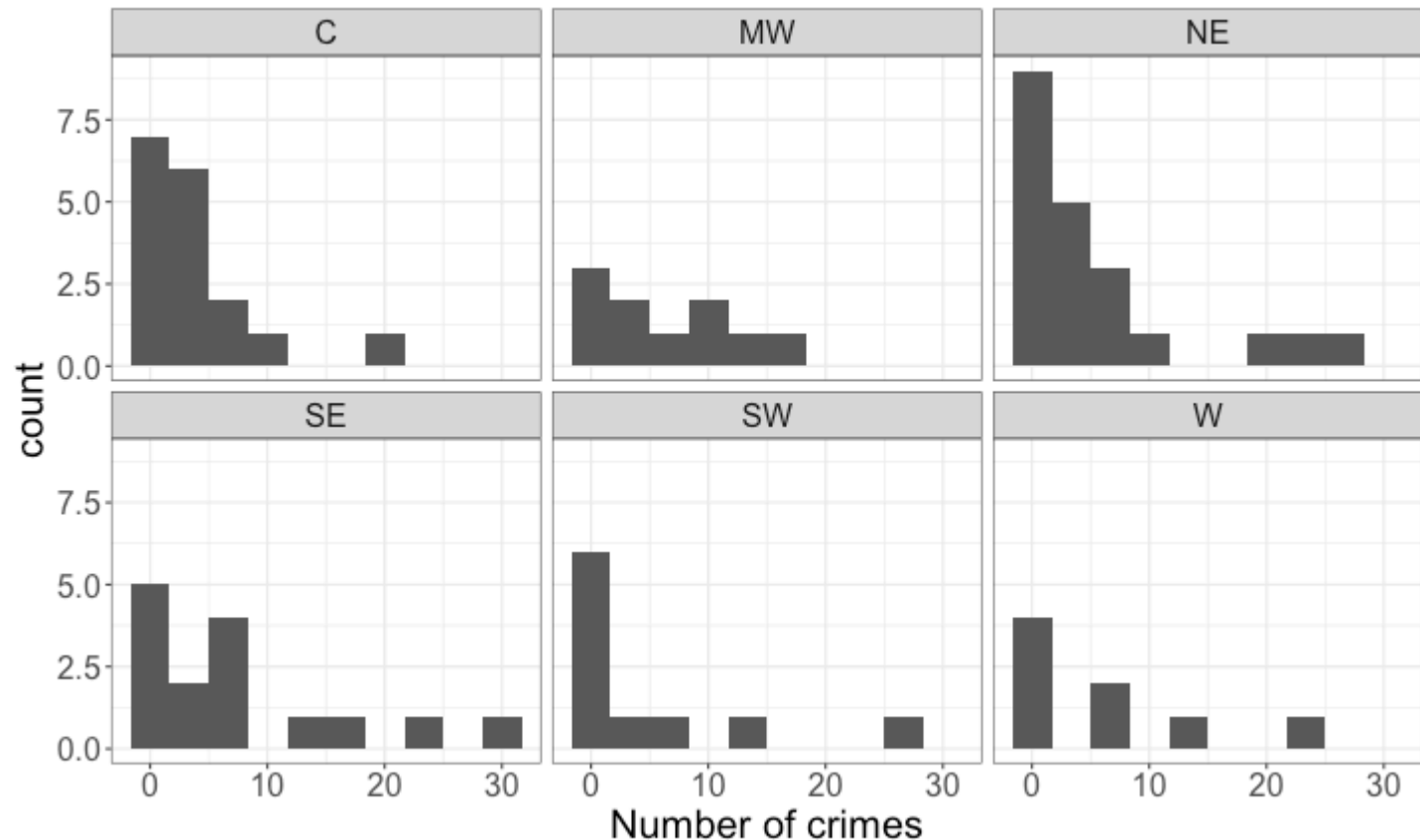
Question

We want to know whether there are regional differences in the number of crimes on college campuses.

$$Crimes_i \sim Poisson(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i$$

Exploratory data analysis



Does it look reasonable to assume a Poisson distribution for the response?

For a Poisson,

$$\text{mean} = \text{variance} = \lambda$$

Exploratory data analysis

```
crimes |>
  group_by(region) |>
  summarize(mean_crimes = mean(nv),
             var_crimes = var(nv))
```

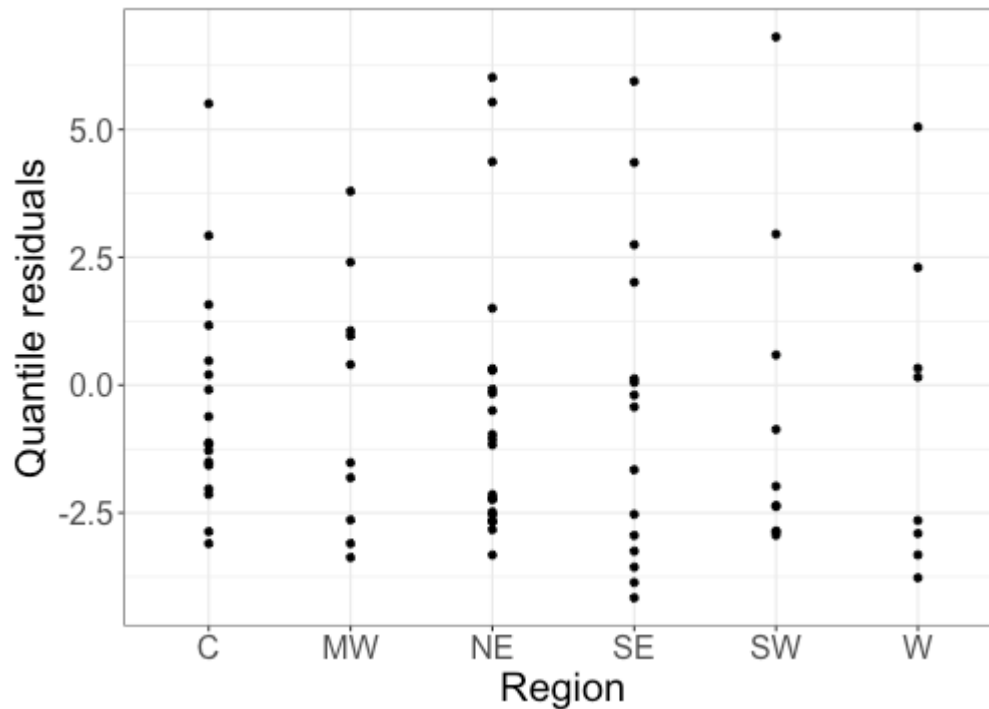
```
## # A tibble: 6 × 3
##   region mean_crimes var_crimes
##   <chr>      <dbl>      <dbl>
## 1 C         3.82        24.3
## 2 MW        6.2         37.1
## 3 NE        5.95        59.0
## 4 SE        8.27        84.4
## 5 SW        5.3         75.3
## 6 W         6.5         65.7
```

Too much
variability!

Does the Poisson distribution still seem reasonable?

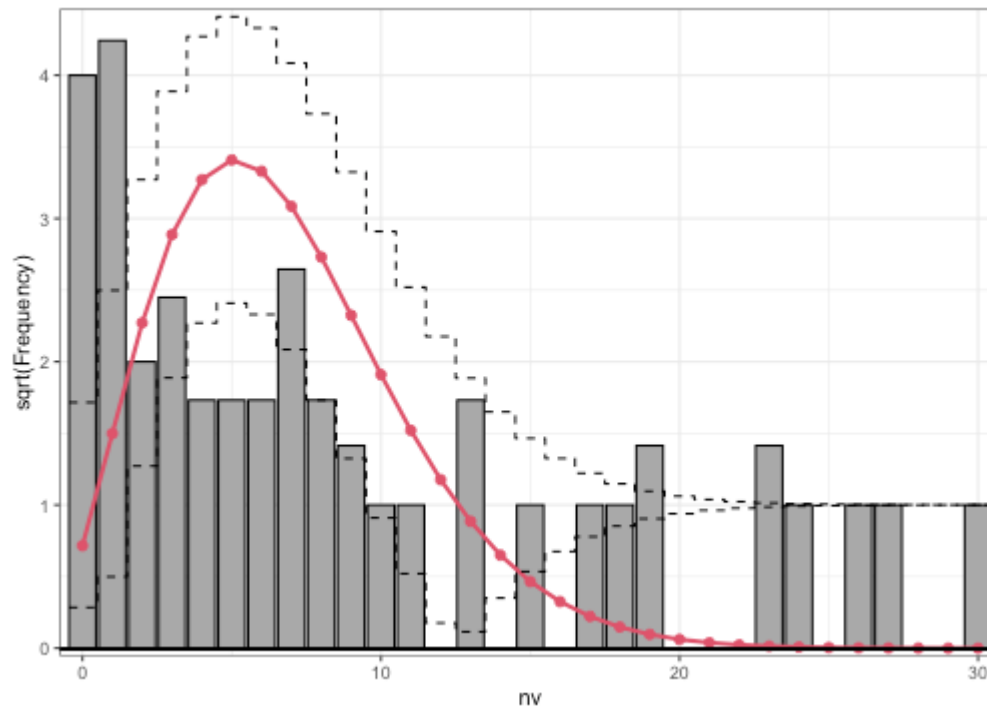
Quantile residual plot

```
m1 <- glm(nv ~ region, data = crimes, family = poisson)
```



Rootogram

```
m1 <- glm(nv ~ region, data = crimes, family = poisson)
rootogram(m1, style="standing")
```



Goodness of fit

Another way to assess whether our model is reasonable is with a *goodness of fit* test.

Goodness of fit test: If the model is a good fit for the data, then the residual deviance follows a χ^2 distribution with the same degrees of freedom as the residual deviance

...

```
## Residual deviance: 621.24    on 75    degrees of freedom
```

```
## AIC: 851.12
```

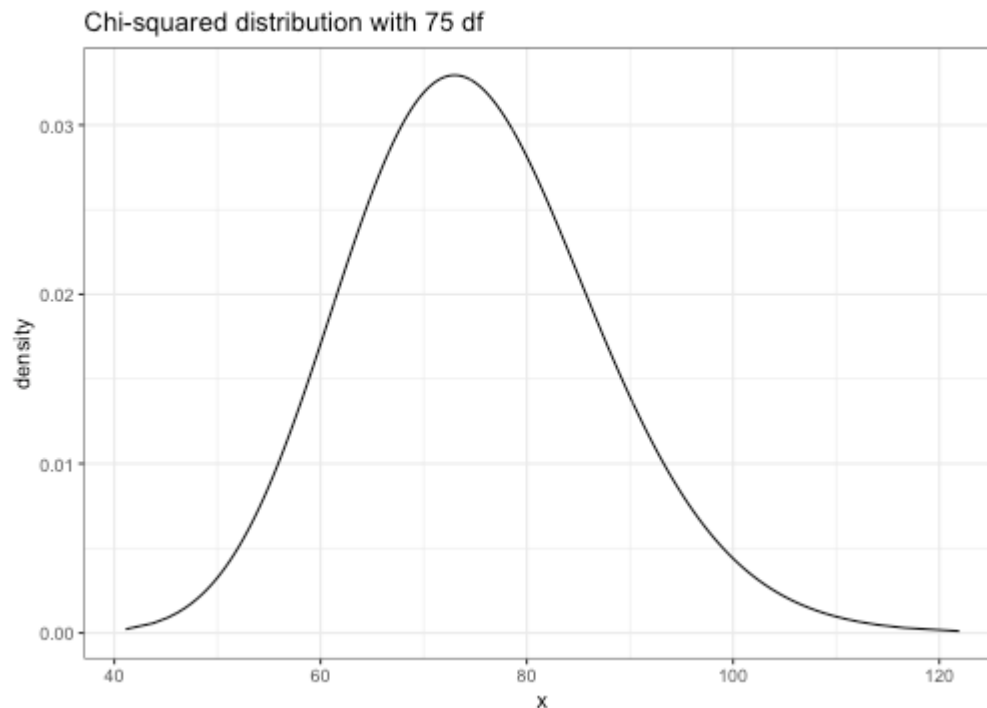
...

Residual deviance = 621.24, df = 75

How likely is a residual deviance of 621.24 if our model is correct?

Goodness of fit

Residual deviance = 621.24, df = 75



```
pchisq(621.24, df=75, lower.tail=F)
```

```
## [1] 5.844298e-87
```

Goodness of fit

Goodness of fit test: If the model is a good fit for the data, then the residual deviance follows a χ^2 distribution with the same degrees of freedom as the residual deviance

Residual deviance = 621.24, df = 75

```
pchisq(621.24, df=75, lower.tail=F)
```

```
## [1] 5.844298e-87
```

So our model might not be a very good fit to the data.

Why might our model not be a good fit?

Potential issues with our model

- + The Poisson distribution might not be a good choice
- + There may be additional factors related to the number of crimes which we are not including in the model

Which other factors might be related to the number of crimes?